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Typical airborne quinones modulate oxidative stress and cytokine expression in lung epithelial A549 cells

Kai Sheng and Jiahuan Lu

Department of Gerontology, Tongren Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

ABSTRACT

Quinones that exist in ambient particulate matter (PM) are hypothesized to be associated with adverse health effects through the generation of reactive oxygen species (ROS). However, the impacts of the quinones on the inflammatory processes have yet to be clearly understood. In this study, we examined the oxidative potentials and biological effects of typical airborne quinones in the human lung epithelial A549 cells. Significant change of redox status, loss of mitochondrial membrane potentials ($\Delta\Psi$) and increase of superoxide dismutase (SOD) activity were induced by exposure to quinones. Some pro-inflammatory genes including interleukin-6 (IL-6), interleukin-8 (IL-8), tumor necrosis factor (TNF- α) and monocyte chemoattractant protein-1 (MCP-1); two aromatic hydrocarbon receptor-regulated genes, cytochromes P450 1A1 (Cyp1a1) and cytochromes P450 1B1 (Cyp1b1); and oxidative stress-related gene heme oxygenase-1 (HO-1) were up-regulated after quinones treatment. Among these quinones, 1,2-naphthoquinone (1,2-NQ) up-regulated expressions of IL-6, IL-8, TNF- α , Cyp1a1, and HO-1; 2-methoxy-1,4-naphthoquinone (MNQ) up-regulated MCP-1, Cyp1b1, Cyp1a1, and HO-1; 2-methylanthraquinone (MAQ) up-regulated IL-6, IL-8, TNF- α , MCP-1, Cyp1b1, and Cyp1a1; acenaphthenequinone (ACQ) up-regulated IL-8, TNF- α , MCP-1, Cyp1b1, and Cyp1a1. These results suggested that all these five quinones had a considerable pro-inflammatory potential by inducing oxidative stress and releasing different types of cytokines/chemokines.

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Ambient particulate matter; cytokines/chemokines; inflammation; oxidative potential; quinones

Introduction

Ambient particulate matter (PM) is an important environmental and occupational health problem in recent years, which is responsible for various cardiovascular events involving cardiovascular mortality and morbidity.^[1–3] The biological effects of PM are mainly due to the soluble organic compounds on the particle surface.^[4,5] However, the underlying mechanisms remain poorly understood because of their complex and heterogeneous properties.

Quinones are one class of the important compounds on the PM surface with potent electrophiles; thus, the nucleophiles attack and the intracellular sulfhydryl arylation may contribute to their cytotoxicity.^[6] Most airborne quinones such as anthraquinone, naphthoquinone, and phenanthrenequinone are oxygenated derivatives of polycyclic aromatic hydrocarbons (PAHs), which are mainly resulted from incomplete combustion process of fossil fuels, or formed via atmospheric photochemical conversions.^[7] Harrison et al.^[8] have measured the levels of PAHs, (oxy) quinones and nitro derivatives at three sites on the coast of Saudi Arabia and Jeddah city, revealing that quinones were longer-life secondary pollutants of PAHs, and their concentrations varied among the sampling sites. Considering the persistence of quinones in the environment and accumulation potential in organism, the toxic effects and toxic mechanism of quinones need to be further studied.

The environmental dosage range of the airborne quinones is reported to be from pg m^{-3} to several ng m^{-3} .^[9] A study in

Birmingham during January 2010 measured quinones dosages in air including the most abundant 9,10-phenanthrenequinone (6.1 ng m^{-3}), 1,2-naphthoquinone (3.0 ng m^{-3}), 1,4-naphthoquinone (2.1 ng m^{-3}), and 2-methylanthraquinone (2.8 ng m^{-3}).^[10] Santos et al.^[11] investigated the total quinones in the practical PM_{10} samples collected on quartz fiber filters with a validated method, finding that the concentrations of quinones ranged from 0.27 to 115 ng m^{-3} . An in vitro study using cultured human lung bronchial epithelial cells with $\text{PM}_{2.5}$ collected from the traffic area in Suwon City, Korea, showed that the organic extract of $\text{PM}_{2.5}$ (including quinones) caused significant DNA breakage and micronucleus formation.^[12] Jean-Jacques et al.^[13] assessed the oxidative potential of PM_4 and $\text{PM}_{2.5}$ collected at different sites of a toll station, finding that quinones were strongly associated with the oxidative potential of particulate matter at the tunnel site.

Mitochondria-mediated abnormal reactive oxygen species (ROS) is widely recognized as an aggravating factor in numerous human diseases.^[14] ROS can directly activate the mitochondrial permeability transition, leading to loss of mitochondrial membrane potential (MMP) and mitochondrial membrane integrity.^[15] Excessive ROS generation and reductant depletion could result in tissue injury and cell death.^[16,17] Jana et al.^[18] highlighted the significance of mitochondrial dysfunction induced by quinone oxidation products of dopamine in the pathogenesis of Parkinson's disease. Shang et al.^[19] demonstrated that

different types of quinones, such as 1,2-naphthoquinone, 2-methylantraquinone, 9,10-phenanthrenequinone, 2-methyl-1,4-naphthoquinone, and acenaphthene-quinone, have various cellular biological responses. ROS formation might be a pivotal mechanism underlying the cytotoxicity. Xia et al.^[20] also reported that disruption of the redox cycling induced by quinones and/or their corresponding semiquinone radicals has been proposed as an important pathway contributing to the cytotoxicity and genotoxicity in the lungs for PM_{2.5}.

The inflammation process is important in the development and exacerbation of pulmonary toxic effects induced by diesel exhaust particles (DEP) and other airborne particulates.^[21,22] Increased production and secretion of cytokines and chemokines are key steps in the inflammatory responses. The amount and combination of cytokines and chemokines released from the pulmonary epithelium can influence both the strength of the inflammatory response and the types of recruited immune cells.^[23–25]

Cytochrome P450 enzymes are a large superfamily of heme-proteins involved in the phase I biotransformation of many endogenous and exogenous compounds. In this family, cytochromes P450 1A (Cyp1a) and cytochromes P450 1B (Cyp1b) are known for their roles in the activation of carcinogens and metabolism of drugs.^[26,27] Aromatic hydrocarbon receptor (AhR) is a ligand-activated transcription factor involved in xenobiotics' metabolism, playing roles in chemically induced carcinogenesis^[28] and participating in cytokine/chemokine release.^[29] AhR regulates the expression of Cyp1a1, 1b1 and 1a2 genes, which encode cytochromes P450 1A1, 1B1 and 1A2, respectively. Differences in AhR affinity can reflect variations of CYP1 enzyme inducibility, which is associated with metabolism of compounds and severity of cytotoxicity.^[30]

A549 cells were chosen in the present study based on previous results, which provided high sensitivity for testing cytotoxicity, DNA damage and ROS responses about ambient particles and quinones.^[19] The toxicologically relevant dosages selected in the present study were based on previous study in our laboratory^[19] and the in vitro experiment conducted by other researchers.^[31] We examined the oxidizing capacity of quinones using a dithiothreitol (DTT)-based assay and investigated the redox status in A549 cells using the superoxide dismutase (SOD) assay. In addition, the mitochondrial membrane potentials ($\Delta\Psi$) and expression of multiple cytokine/chemokine genes were examined to explore the underlying mechanism.

Materials and methods

Materials

The A549 cell line was purchased from American Tissue Type Culture Collection (Manassas, VA, USA). RPMI 1640 medium and fetal bovine serum (FBS) were obtained from Invitrogen (Carlsbad, CA, USA). Superoxide dismutase (SOD) assay kit was from Nanjing Jiancheng Bioengineering Research Institute (Nanjing, China). Dimethyl sulfoxide (DMSO), Rhodamine 123 (Rh123), and 5, 5'-dithiobis-2-nitrobenzoic acid (DTNB) were purchased from Sigma (St. Louis, MO, USA). 1,2-Naphthoquinone (1,2-NQ), 2-methoxy-1,4-naphthoquinone (MNQ), 2-

methylantra-quinone (MAQ), acenaphthenequinone (ACQ), and 9,10-phenanthrenequinone (PQ) were purchased from Sigma (USA). TRIzol reagent was purchased from Invitrogen (CA, USA). ReverTra Ace[®] qPCR RT kit and THUNDERBIRD SYBR[®] qPCR mix were purchased from TOYOBO (Japan). All other reagents used were analytical grade chemicals from Sino-pharm Chemical Reagent Company (Shanghai, China).

Cell culture and treatment

A549 cells were cultured in RPMI 1640 medium supplemented with 10% FBS and penicillin/streptomycin (100 units/mL) in an atmosphere of 5% CO₂ and 100% relative humidity at 37°C. Quinones were dissolved in DMSO, and the working solutions were freshly diluted by the RPMI 1640 with 2% FBS. The final concentration of DMSO was 0.1% (v/v). Treatment dosages were 5, 10, 15, 20 μ M for 1, 2-NQ and MNQ; 10, 20, 40, 60 μ M for MAQ and ACQ; 1, 2, 3, 4 μ M for PQ. Cells were co-incubated with quinones for 4 h except for the SOD detection experiment (incubation for 24 h). DMSO-treated cells (0.1%, v/v) served as the control.

Dithiothreitol (DTT) assay

DTT assay, a chemical-based redox-active assay, is used to quantify the catalytically active redox species. In this study, all spectrophotometric measurements were performed using a Multiskan Mk3 plate reader (Thermo Electron Corporation, Waltham, MA, USA). We followed the procedure of Cho et al.^[32] for DTT experiment. Briefly, 10 μ M of quinones were incubated at 37°C with DTT (100 μ M) in 0.10 M phosphate buffer at pH 7.4 for different time intervals (PQ varying from 0 to 10 min, 1, 2-NQ from 0 to 20 min, MNQ from 15 to 180 min). At designated time intervals, a 0.5 mL aliquot of the reaction mixture was removed, and 0.5 mL of 10% trichloroacetic acid was added to stop the reaction. Finally, 50 μ L of 10.0 mM DTNB (made in 0.1 M phosphate buffer, pH 7.4) was added and allowed to react for 5 min; then, 2.0 mL of 0.4 M Tris-buffer (0.4 M Tris HCl, pH 8.9 in 20 mM EDTA) was added. Light absorption of the product (2-nitro-5-mercapto-benzoic acid; TNB) formed by the reaction of DTNB with residual DTT was measured at 412 nm. The DTT assay response (R_{DTT}) was calculated as follows:

$$R_{\text{DTT}} (\%) = (\text{OD}_{\text{sample}} - \text{OD}_{\text{blank}}) / \text{OD}_{\text{blank}} \times 100.$$

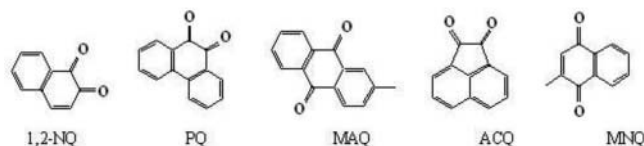
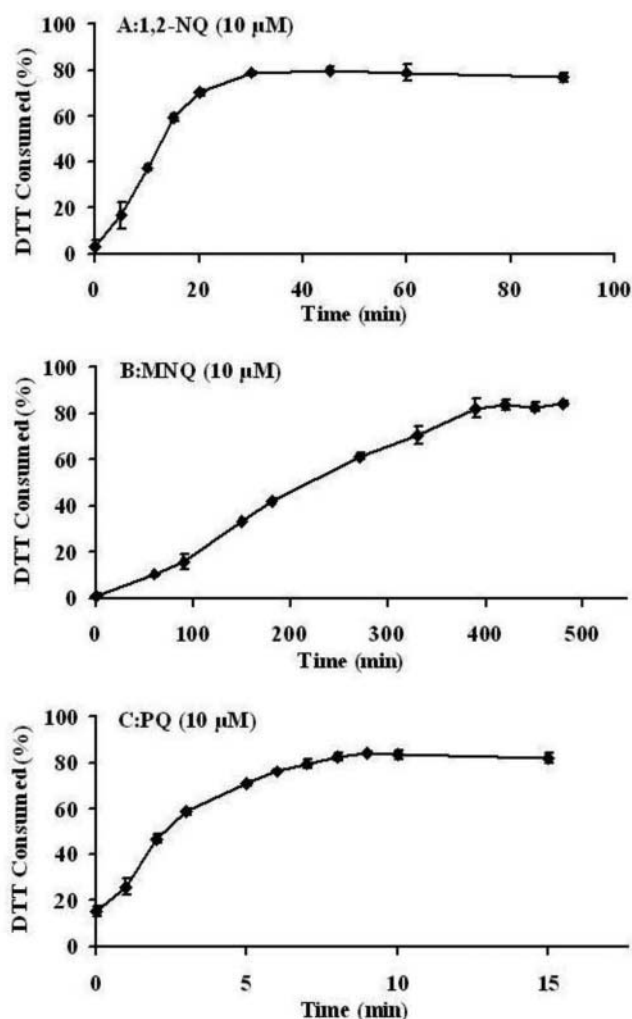
Superoxide dismutase (SOD) assay

SOD is a family of antioxidative enzymes that act as the first cell defense against oxidative stress, catalyzing the dismutation of O₂⁻ to H₂O₂.^[33] The SOD activity was measured according to the manufacturer's procedure. Briefly, A549 cells were plated in the culture flask at a density of 1×10^7 cells per flask and cultured for 24 h to allow cell adhesion. Then, the culture medium was replaced with fresh medium containing quinones. After reaction for 24 h, the SOD activity was assayed by colorimetry. The OD was recorded using a Quant microplate spectrophotometer (BioTek Instruments, Winooski, VT) at a

Table 1. The forward and reverse primers of RT-PCR used to amplify the selected genes.

Gene	Forward primer	Reverse primer
CYP1A1	5'-ACTTCATCCCTATTCTCGCTACCT-3'	5'-CGGATGTGGCCCTTCTCA-3'
CYP1B1	5'-GTACCGGCCATCATCACTGACA-3'	5'-CACATCAGGATACCTGGTGAAGAG-3'
IL6	5'-CTTCTCCACAAGCGCTTC-3'	5'-GGCAAGTCTCCTCATTGAATC-3'
IL8	5'-AACCCCTCTGCACCCAGTTTCTCT-3'	5'-TCCACTCTCAATCACTCTCA-3'
TNF α	5'-CAGGCAGTCAGATCATCTTC-3'	5'-CTTGATGGCAGAGAGGAGGT-3'
Mcp-1	5'-TCCAGCATGAAAGTCTCTGC-3'	5'-TGG AATCCTGAACCCACTTC-3'
HO-1	5'-AAGACTGCGTTCCTGCTCAAC-3'	5'-AAAGCCCTACAGCAACTGTCTG-3'
GAPDH	5'-CCATGGAGAAGGCTGGG-3'	5'-CAAAGTTGTCATGGATGACC-3'

Cyp1a1: cytochromes P450 1A1; Cyp1b1: cytochromes P450 1B1; IL-6: interleukin-6; IL-8: interleukin-8; TNF- α : tumor necrosis factor; Mcp-1: monocyte chemotactic protein 1; HO-1: heme oxygenase-1; GAPDH: glyceraldehyde-3-phosphate dehydrogenase.

**Figure 1.** Chemical structures of 1,2-NQ, PQ, MAQ, ACQ, and MNQ.**Figure 2.** Quinones-induced DTT consumption. DTT (100 μ M) was incubated with 10 μ M of 1,2-NQ (A), MNQ (B), or PQ (C) at 37°C in 0.1 M phosphate buffer (pH 7.4) for different time. Experiments were repeated three times with three duplicates for each sample. 1,2-NQ: 1,2-naphthoquinone; PQ: 9,10-phenanthrenequinone; MNQ: 2-methoxy-1,4-naphthoquinone.

wavelength of 550 nm, and the resulting data were calculated as follows: SOD activity (U/mg prot) = [(ODcontrol-ODsample)/ODcontrol]/50% $\times$ (reactive volume /sample volume) $\times$ dilution factor/sample protein concentrations.

Assay of mitochondrial membrane potential ($\Delta\Psi_m$)

To evaluate the effect of quinones on mitochondrial membrane potential, we used a fluorescent dye Rh123, which can aggregate in the mitochondria and emit yellow green fluorescence. The uptake of Rh123 into the mitochondria is dependent on mitochondrial membrane potential. Briefly, A549 cells were plated in 35-mm culture dishes at a density of 2×10^4 cells per dish and cultured for 24 h to allow cell adhesion. Subsequently, cells were washed twice with warm D-hank's solution, incubated with Rh123 at 37°C for 30 min in the dark, and observed under a fluorescence microscope (Olympus BX-51; Olympus, Japan). The intensity of fluorescence was analyzed by Image-Pro Plus 6, and the fluorescence of the control group was normalized to 100%.

Quantitative real-time polymerase chain reaction (qRT-PCR)

For PCR arrays, A549 cells were seeded in six-well culture plates (5×10^5 /well) and cultured for 24 h. Cells were exposed to quinones and incubated for 4 h at 37°C. Total RNA was isolated from the cells using the high pure RNA isolation kit (Toyobo, Japan). The RNA concentration was determined spectrometrically using the Gene Pulser Xcell Total System (Bio-Rad, Hercules, CA, USA). Total RNA (1 μ g) was used for the first-strand cDNA reaction using reverse transcriptase (Toyobo, Japan). Expression levels for each cDNA sample were quantified with the Thunderbird SYBR[®] qPCR mix (Toyobo, Japan) using the Exicycler[™] 96 system (Bioneer, Korea). Seven genes including Cyp1a1 (cytochrome P450, family 1, subfamily a, polypeptide 1), Cyp1b1 (cytochrome P450, family 1, subfamily b, polypeptide 1), IL-6 (interleukin-6), IL-8 (interleukin-8), TNF- α (tumor necrosis factor), Mcp-1/Ccl2 (monocyte chemotactic protein 1, and HO-1 (heme oxygenase-1) were analyzed. The following primer sequences were used in qRT-PCR (Table 1).

The expression of each gene of interest was normalized against the reference gene (glyceraldehyde-3-phosphate dehydrogenase, GAPDH) and was expressed as fold change compared with the control. The fold change was calculated by the $\Delta\Delta C_t$ method according to the following formula: $\Delta\Delta C_t =$

$Ct_{(G0I)} - Ct_{(Gapdh)}$; $\Delta\Delta Ct = \Delta Ct_{(treated)} - \Delta Ct_{(control)}$; Induction fold = $2^{(-\Delta\Delta Ct)}$

Data analysis

Data were analyzed with SPSS software (ver. 16.0). All data were expressed as mean $\pm$ standard error of the mean (SEM) and were compared using one-way analysis of variance (ANOVA). In all cases, $P < 0.05$ was considered statistically significant.

Results

Oxidative potential

The chemical structures of these five quinones are shown in Figure 1. The time-dependent consumption of DTT catalyzed by quinones is shown in Figure 2. This experiment was carried out by adding quinones with a final concentration of 10 μ M into the buffer solution containing DTT and incubating at 37°C for different time intervals varying from 0 to 480 min. The time-dependent plot is linear when the DTT consumption is less than 85%. Under the linear ranges, the DTT oxidation rates for 1,2-NQ, MNQ, and PQ were 2.73 min⁻¹, 0.21 min⁻¹, and 7.80 min⁻¹, respectively. At 10 μ M of concentration, the DTT was not consumed by MAQ and ACQ (data not shown).

SOD activity

To investigate whether quinones increase the SOD activity in A549 cells, we measured the release of SOD enzyme in A549 cells. As shown in Figure 3a and b, 1, 2-NQ (5 μ M) and MNQ (5 μ M) caused a significant increase in SOD activity, which was 2.52 and 1.90 times higher over the control groups, respectively. However, over 10 μ M of 1, 2-NQ and MNQ caused a dose-dependent decline of SOD activity. In addition, the SOD activity in the 20 μ M 1, 2-NQ/MNQ groups was almost equal to that in the control groups. MAQ, ACQ, and PQ increased the SOD activities in a dose-dependent manner in the dose range of this study (ANOVA test, $P < 0.01$) (Fig. 3c–e), with the uppermost SOD activity being 4.39, 2.59, and 11.14 times higher than that of control group, respectively.

Mitochondrial membrane potential (MMP)

The rhodamine 123 staining assay was used to investigate the mitochondrial membrane potential in A549 cells treated with quinones for 4 h. As shown in Figure 4a, c, and e, 1,2-NQ, MAQ, and PQ induced obvious loss of mitochondrial membrane potential (MMP) in a dose-dependent manner (ANOVA test, $P < 0.01$). At the maximum concentration of 1,2-NQ (20 μ M), MAQ (60 μ M), and PQ (4 μ M), the MMP reduced to 22.25%, 74.02%, and 41.86% as compared with the control groups ($P < 0.05$). MNQ and ACQ seemed to have no effect on the MMP (Fig. 4b and d).

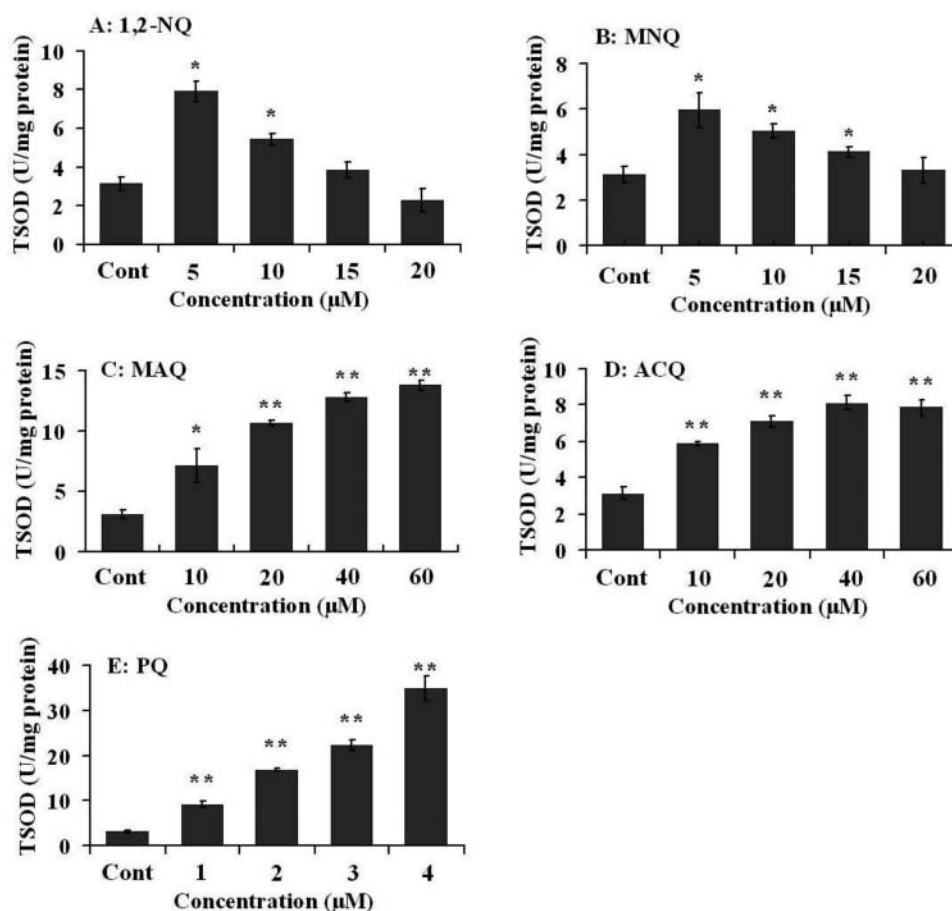


Figure 3. Effect of 1,2-NQ (A), MNQ (B), MAQ (C), ACQ (D), and PQ (E) on the SOD activity in A549 cells after 24 h treatment. The OD was recorded using a μ Quant microplate spectrophotometer at a wavelength of 550 nm. All values are expressed as mean $\pm$ SEM. Experiments were repeated three times with three duplicates for each sample. Control was cells treated with 0.1% DMSO (v/v) only. * $P < 0.05$ and ** $P < 0.01$, compared with control.

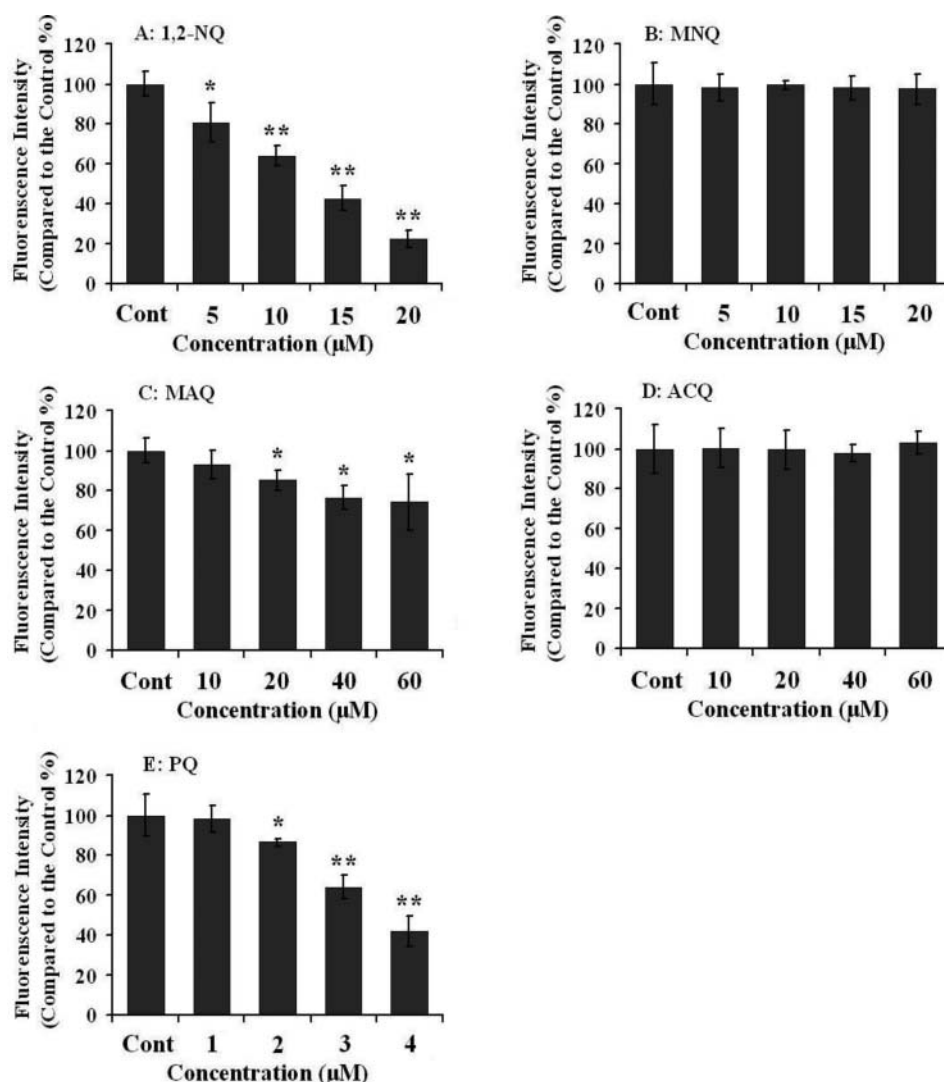


Figure 4. Effect of 1,2-NQ (A), MNQ (B), MAQ (C), ACQ (D), and PQ (E) on MMP reduction in A549 cells. The rhodamine 123 staining assay was used to investigate the mitochondrial membrane potential in A549 cells treated with quinones for 4 h. All values were expressed as mean $\pm$ SEM. Experiments were repeated three times with three duplicates in each sample. Control cells were treated with 0.1% DMSO (v/v) only. * $P < 0.05$ and ** $P < 0.01$, compared with control.

Cytokine/chemokine expressions

Cytokine and chemokine induction is a pivotal step in the inflammatory process. The type and amount of cytokine and chemokine determine the outcome of the pro-inflammatory responses. In this study, we investigated the effects of quinones on the expressions of following four cytokine/chemokine genes in A549 cells: IL-6, IL-8, TNF- α , and MCP-1. As shown in Figure 5 and Table 2, IL-8, TNF- α , and IL-6 were observed to be up-regulated in A549 cells treated with 1, 2-NQ and MAQ. Among these cytokines, the expression level of IL-8 increased in a dose-dependent manner. In addition, the up-regulation of TNF- α gene expression was observed only in 20 μ M group of 1,2-NQ-treated A549 cells. The expression of MCP-1 was also significantly up-regulated in MAQ treatment groups (10, 20, 40, and 60 μ M), but not in 1, 2-NQ treatment groups. In the MNQ-treated groups, obvious induction of gene expression was observed only for MCP-1. ACQ treatment induced significant expression of IL-8, TNF- α , and MCP-1 ($P < 0.05$, ANOVA test), but not IL-6. In the PQ-treated A549 cells, the expressions

of IL-8, TNF- α , IL-6 and MCP-1 gene were not changed. The summary of gene expression was described in Table 2.

In addition, the expression of two AhR-regulated genes, CYP1a1 and CYP1b1 (encoding cytochromes P450 1A1 and 1B1, respectively), was also measured in A549 cells exposed to quinones (Fig. 5, Table 2). CYP1a1 and CYP1b1 genes were up-regulated in A549 cells after treatment with MNQ, MAQ, and ACQ. However, 1, 2-NQ could induce the expression of CYP1a1 (but not CYP1b1) only at 5 μ M concentration group. PQ exposure induced neither CYP1a1 nor CYP1b1 gene expression in A549 cells. In addition, 1,2-NQ, MNQ, and PQ could significantly induce the expression of HO-1 (especially 1, 2-NQ), while MAQ and ACQ did not induce the HO-1 expression.

Discussion

The nuclear magnetic resonance (NMR) analysis confirmed that quinones moieties were in the humic substances, which can serve as important reversible redox sites.^[34,35] It is well

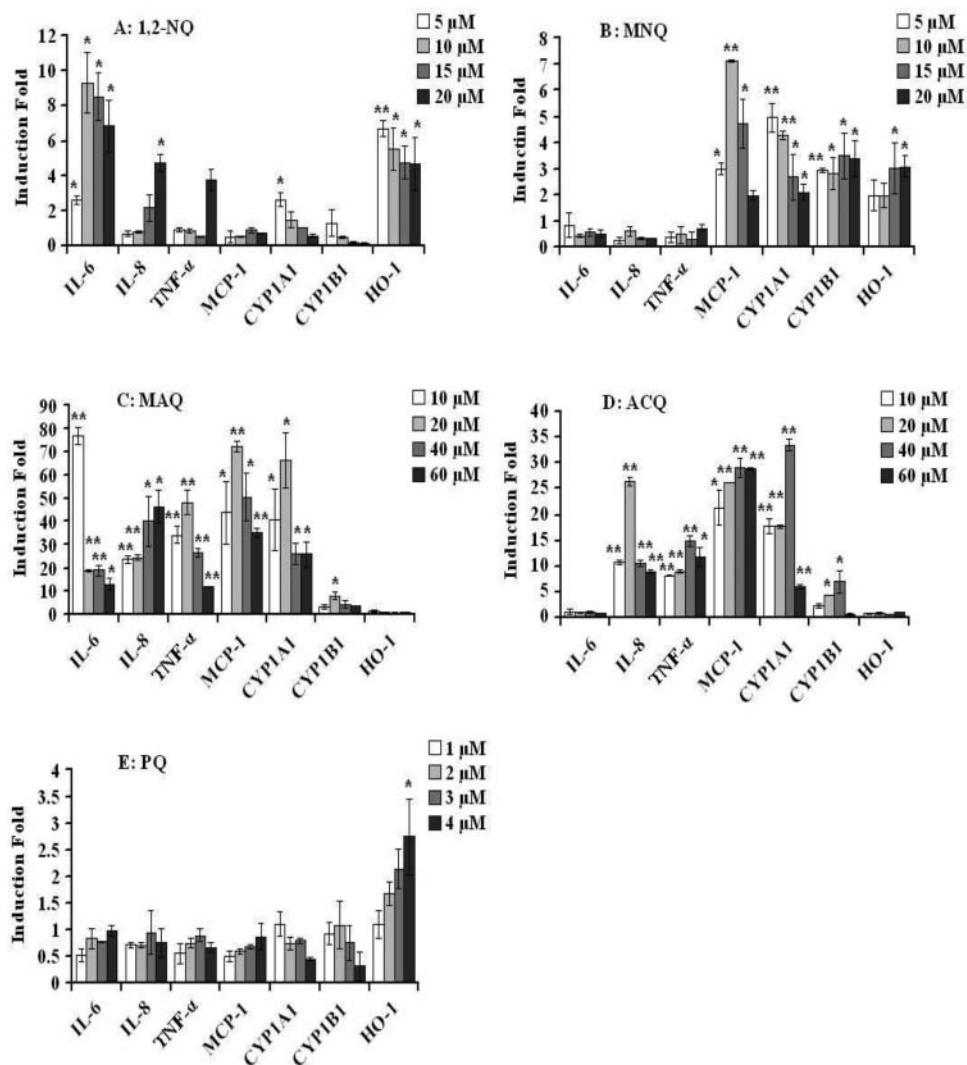


Figure 5. Gene expressions as measured by qRT-PCR in A549 cells treated with 1,2-NQ (A), MNQ (B), MAQ (C), ACQ (D), and PQ (E) for 4 h. Data were presented as induction fold as compared with DMSO-treated (0.1%) control group and were normalized to GAPDH mRNA levels. Each bar represents the average of three independent experiments with three replicates for each sample. * $P < 0.05$ and ** $P < 0.01$, compared with control.

known that quinones can effectively catalyze the electron transfer reactions due to their ability to form stable semi-quinone radicals, which can either obtain one electron from reductant (e.g., DTT) or donate one electron to oxidant (e.g., O_2). In this study, the oxidative potential of the quinones was measured by the DTT assay. This assay monitors the ability of quinones to catalyze the reduction of oxygen (O_2) to superoxide radical by DTT. The DTT assay of the present study showed that PQ, 1,2-NQ, and MNQ caused significant DTT consumption, while MAQ and ACQ did not react with DTT, indicating the

oxidation-induction ability of MAQ and ACQ was comparatively weak. This result is consistent with that of other reports that ROS could be induced by five quinones, while MAQ and ACQ generated ROS only at high concentration.^[19]

It has been well established that SOD provides an important defense mechanism against superoxide anion by detoxifying the ROS and modulating the redox-sensitive cell-signaling pathways. Upon challenge with low levels of free radicals, the SOD activity can be mobilized via various initial adaptive pathways. Therefore, the cellular SOD activity can indirectly

Table 2. Induced expression of the selected genes after quinones treatment.

Max folds	IL-6	IL-8	TNF α	Mcp-1	CYP1A1	CYP1B1	HO-1
1, 2-NQ	9.31 $\pm$ 1.73*	4.74 $\pm$ 0.45*	3.76 $\pm$ 0.61*	0.48 $\pm$ 0.32*	2.61 $\pm$ 0.41*	0.18 $\pm$ 0.04	6.70 $\pm$ 0.44**
MNQ	0.41 $\pm$ 0.06	0.22 $\pm$ 0.11	0.27 $\pm$ 0.27	7.11 $\pm$ 0.03**	4.95 $\pm$ 0.55**	3.48 $\pm$ 0.88*	3.08 $\pm$ 0.43*
MAQ	76.71 $\pm$ 3.49**	46.35 $\pm$ 7.04*	48.12 $\pm$ 5.40**	72.33 $\pm$ 2.13**	65.98 $\pm$ 11.92*	7.96 $\pm$ 1.70*	0.45 $\pm$ 0.27
ACQ	0.81 $\pm$ 0.07	26.47 $\pm$ 0.81**	14.87 $\pm$ 1.11**	29.08 $\pm$ 1.89**	33.48 $\pm$ 1.08**	6.93 $\pm$ 2.20*	0.56 $\pm$ 0.05
PQ	0.52 $\pm$ 0.11	0.70 $\pm$ 0.04	0.54 $\pm$ 0.18	0.49 $\pm$ 0.09	0.45 $\pm$ 0.03	0.30 $\pm$ 0.27	2.74 $\pm$ 0.70*

1,2-NQ, 1,2-Naphthoquinone; MNQ, 2-Methoxy-1,4-naphthoquinone; MAQ, 2-Methylanthra-quinone; ACQ, Acenaphthenequinone; PQ, 9,10-phenanthrenequinone.

* $P < 0.05$ and ** $P < 0.01$, compared with control.

indicate the toxicant-induced ROS level. In this study, quinones elevated the SOD activity as compared with the DMSO-treated group ($P < 0.05$) in A549 cells, suggesting that the quinones could efficiently induce ROS. A similar response of ROS formation was observed in previously published studies.^[36] With the increase of 1,2-NQ and MNQ concentration, however, excessive SOD mobilization and consumption might lead to depletion of SOD, as was observed in the present study. MAQ and ACQ could only induce slight increase of SOD activity, suggesting that the oxidative stress effect of MAQ and ACQ were relatively weak, which was consistent with the results of DTT assay.

Mitochondria are the major source of ROS generation, which is mediated by permeability transition.^[37] It has been reported that mitochondrial derived ROS could synergistically regulate stochastic mitochondrial permeability transition pore (mPTP) opening and quantal bursting mode of ROS production in many types of cells.^[38] Mitochondrial dysfunction has been demonstrated to participate in the induction of cell death through release of mitochondrial apoptogenic factors and a decrease of ATP generation.^[39–41] Our study revealed that quinones (especially PQ, 1,2-NQ, MNQ) induced increase of ROS and loss of mitochondrial membrane potential. These findings suggested that mitochondrial derived ROS generation may be an important pathway involved in the toxic mechanism of quinones, which was consistent with the report of Shang et al.^[19] that quinones induced cytotoxicity through generation of ROS.

Pro-inflammatory response is central to the development of various health effects associated with airborne particles.^[22] In the present study, we demonstrated that five quinones exerted extensive pro-inflammatory effects in A549 cells. The differences in cytokine/chemokine profiles induced by the tested compounds suggested that they might lead to activation of different groups of immune cells. IL-6 was induced by 1,2-NQ and MAQ in this study, which is well known as a cancer progression-related cytokine.^[42] IL-8, induced by 1,2-NQ, MAQ and ACQ, is predominantly a neutrophil-recruiting chemokine associated with pulmonary edema, neutrophilia, as well as broncho-constriction.^[43,44] MAQ and ACQ induced cytokine secretion of both TNF- α and MCP-1 at the same time. These findings suggested that the five quinones might potentially be important in the inflammation process or carcinogenic potency.

The expression of CYP1 genes is regulated by aryl hydrocarbon receptor (AhR), which forms an active transcription factor heterodimer with the AhR nuclear translocator (ARNT) after ligand-binding. It was reported that the aryl hydrocarbon receptor (AhR) could bind with nitro-PAHs, resulting in up-regulation of AhR-regulated genes such as CYP enzymes.^[45] CYP1A1 and CYP1B1 are involved in activating PAHs to form cancer-causing reactive intermediates, which can subsequently lead to replication errors or mutations, if not repaired in time. In the present study, the real-time PCR results showed that MNQ, ACQ and MAQ could significantly induce the expression of both CYP1A1 and CYP1B1. PQ failed to up-regulate CYP1A1 or CYP1B1, while 1,2-NQ induced expression of CYP1A1 at low concentration (5 μ M).

Heme oxygenase (HO-1) is an inducible isoform of the first and rate-controlling enzyme in the catabolism of heme, catalyzing heme into biliverdin, free iron and carbon monoxide. HO-1

has anti-inflammatory and anti-apoptotic capability, playing critical roles in defending cells against oxidant-induced injury. Choi and Alam^[46] reported that the biological significance of HO-1 expression in the lung was the antioxidant effect of its catabolic product. HO-1 pathway acts as an important protective endogenous mechanism against physical, chemical, and biological stress.^[47] In this study, we demonstrated that MNQ and PQ and 1, 2-NQ up-regulated the expression of HO-1, which was consistent with the results of SOD activity.

Conclusion

The present study found that exposure to quinones caused oxidative stress response and induced expression of various cytokine/chemokine in A549 cells, suggesting that quinones might be an important part of the PAH subgroup, contributing to the toxicity of airborne particulate matter. Further studies on the long term effects of these contaminants and the full scope of possible toxic effects should be investigated to clarify the underlying mechanism.

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Conflict of interest statement

The authors declare that there are no conflicts of interest.

References

- [1] Wichmann, H.E. Diesel exhaust particles. *Inhal. Toxicol.* **2007**, *19*, 241–244.
- [2] Bell, M.L.; Ebisu, K.; Peng, R.D.; Samet, J.M.; Dominici, F. Hospital admissions and chemical composition of fine particle air pollution. *Am. J. Respir. Crit. Care Med.* **2009**, *179*, 1115–1120.
- [3] Peng, R.D.; Chang, H.H.; Bell, M.L.; McDermott, A.; Zeger, S.L.; Samet, J.M.; Dominici, F. Coarse particulate matter air pollution and hospital admissions for cardiovascular and respiratory diseases among medicare patients. *JAMA.* **2008**, *299*, 2172–2179.
- [4] Campen, M.J.; Babu, N.S.; Helms, G.A.; Pett, S.; Wernly, J.; Mehran, R.; McDonald, J.D. Nonparticulate components of diesel exhaust promote constriction in coronary arteries from ApoE $^{-/-}$ mice. *Toxicol. Sci.* **2005**, *88*, 95–102.
- [5] Yokota, S.; Ohara, N.; Kobayashi, T. The effects of organic extract of diesel exhaust particles on ischemia/reperfusion-related arrhythmia and on pulmonary inflammation. *Toxicol. Sci.* **2008**, *33*, 1–10.
- [6] Gant, T.W.; Rao, D.N.; Mason, R.P.; Cohen, G.M. Redox cycling and sulphhydryl arylation; their relative importance in the mechanism of quinone cytotoxicity to isolated hepatocytes. *Chem. Biol. Interact.* **1998**, *65*, 157–173.
- [7] Walgraeve, C.; Demeestere, K.; Dewulf, J.; Zimmermann, R.; Langenhove, H.V. Oxygenated polycyclic aromatic hydrocarbons in atmospheric particulate matter: Molecular characterization and occurrence. *Atmos. Environ.* **2010**, *44*, 1831–1846.
- [8] Harrison, R.M.; Alam, M.S.; Dang, J.; Ismail, I.M.; Basahi, J.; Alghamdi, M.A.; Hassan, I.A.; Khoder, M. Relationship of polycyclic aromatic hydrocarbons with oxy(quinone) and nitro derivatives during air mass transport. *Sci. Total Environ.* **2016**, *S0048-9697(16)*, 31722–31723.
- [9] Walgraeve, C.; Demeestere, K.; Dewulf, J.; Zimmermann, R.; Langenhove, H.V. Oxygenated polycyclic aromatic hydrocarbons in

- atmospheric particulate matter: Molecular characterization and occurrence. *Atmos. Environ.* **2010**, *44*, 1831–1846.
- [10] Alam, M.S.; Delgado-Saborit, J.M.; Stark, C.; Harrison, R.M. Using atmospheric measurements of PAH and quinone compounds at roadside and urban background sites to assess sources and reactivity. *Atmos. Environ.* **2013**, *77*, 24–35.
- [11] Santos, A.G.; Regis, A.C.; da Rocha, G.O.; Bezerra Mde, A.; de Jesus, R.M.; de Andrade, J.B. A simple, comprehensive, and miniaturized solvent extraction method for determination of particulate-phase polycyclic aromatic compounds in air. *J Chromatogr. A* **2016**, *1435*, 16–17.
- [12] Oh, S.M.; Kim, H.R.; Park, Y.J.; Lee, S.Y.; Chung, K.H. Organic extracts of urban air pollution particulate matter (PM_{2.5})-induced genotoxicity and oxidative stress in human lung bronchial epithelial cells (BEAS-2B cells). *Mutat. Res.* **2011**, *723*(2), 142–151.
- [13] Jean-Jacques, S.; Simon, D.; Ferdinand, S.; Michael, R. Oxidative potential of particles in different occupational environments: A pilot study. *Ann. Occup. Hyg.* **2015**, *59*(7), 882–894.
- [14] Yadav, N.; Chandra, D. Mitochondrial and postmitochondrial survival signaling in cancer. *Mitochondrion* **2014**, *16*, 18–25.
- [15] Crompton, M.; Barksby, E.; Johnson, N.; Capano, M. Mitochondrial intermembrane junctional complexes and their involvement in cell death. *Biochimie* **2002**, *84*, 143–152.
- [16] Kang, J.; Pervaiz, S. Mitochondria: Redox metabolism and dysfunction. *Biochem. Res. Int.* **2012**, *2012*, 896751–896765.
- [17] Yoboue, E.D.; Devin, A. Reactive oxygen species-mediated control of mitochondrial biogenesis. *Int. J. Cell Biol.* **2012**, *2012*, 403870–403877.
- [18] Jana, S.; Sinha, M.; Chanda, D.; Roy, T.; Banerjee, K.; Munshi, S.; Patro, B.S.; Chakrabarti, S. Mitochondrial dysfunction mediated by quinone oxidation products of dopamine: Implications in dopamine cytotoxicity and pathogenesis of Parkinson's disease. *Biochim. Biophys. Acta* **2011**, *1812*(6), 663–673.
- [19] Shang, Y.; Zhang, L.; Jiang, Y.; Li, Y.; Lu, P. Airborne quinones induce cytotoxicity and DNA damage in human lung epithelial A549 cells: The role of reactive oxygen species. *Chemosphere* **2014**, *100*, 42–49.
- [20] Xia, T.; Korge, P.; Weiss, J.N.; Li, N.; Venkatesen, M.I.; Sioutas, C.; Nel, A. Quinones and aromatic chemical compounds in particulate matter induce mitochondrial dysfunction: Implications for ultrafine particle toxicity. *Environ. Health Perspect.* **2004**, *112*(14), 1347–1358.
- [21] Donaldson, K.; Stone, V. Current hypotheses on the mechanisms of toxicity of ultrafine particles. *Ann. Ist. Super. Sanita* **2003**, *39*, 405–410.
- [22] Salvi, S.; Holgate, S.T. Mechanisms of particulate matter toxicity. *Clin. Exp. Allergy* **1999**, *29*, 1187–1194.
- [23] Baggiolini, M. Chemokines in pathology and medicine. *J. Intern. Med.* **2001**, *250*, 91–104.
- [24] Bisset, L.R.; Schmid-Grendelmeier, P. Chemokines and their receptors in the pathogenesis of allergic asthma: Progress and perspective. *Curr. Opin. Pulm. Med.* **2005**, *11*, 35–42.
- [25] Holtzman, M.J.; Morton, J.D.; Shornick, L.P.; Tyner, J.W.; O'Sullivan, M.P.; Antao, A.; Lo, M.; Castro, M.; Walter, M.J. Immunity, inflammation, and remodeling in the airway epithelial barrier: Epithelial-viral-allergic paradigm. *Physiol. Rev.* **2002**, *82*, 19–46.
- [26] Anzenbacher, P.; Anzenbacherova, E. Cytochromes P450 and metabolism of xenobiotics. *Cell. Mol. Life Sci.* **2001**, *58*, 737–747.
- [27] Monostory, K.; Pascucci, J.M.; Kobori, L.; Dvorak, Z. Hormonal regulation of CYP1A expression. *Drug Metab. Rev.* **2009**, *41*, 547–572.
- [28] Abel, J.; Haarmann-Stemmann, T. An introduction to the molecular basics of aryl hydrocarbon receptor biology. *Biol. Chem.* **2010**, *391*, 1235–1248.
- [29] Haarmann-Stemmann, T.; Bothe, H.; Abel, J. Growth factors, cytokines and their receptors as downstream targets of arylhydrocarbon receptor (AhR) signaling pathways. *Biochem. Pharmacol.* **2009**, *77*, 508–520.
- [30] Nebert, D.W.; Dalton, T.P.; Okey, A.B.; Gonzalez, F.J. Role of aryl hydrocarbon receptor-mediated induction of the CYP1 enzymes in environmental toxicity and cancer. *J. Biol. Chem.* **2004**, *279*, 23847–23850.
- [31] Carneiro, P.F.; Pinto, M.C.; Marra, R.K.; da Silva Fde, C.; Resende, J. A.; Rocha, E.; Silva, L.F.; Alves, H.G.; Barbosa, G.S.; de Vasconcellos, M.C.; Lima, E.S.; Pohlit, A.M.; Ferreira, V.F. Synthesis and antimalarial activity of quinones and structurally-related oxirane derivatives. *Eur. J. Med. Chem.* **2016**, *108*, 134–140.
- [32] Cho, A.K.; Sioutas, C.; Miguel, A.H.; Kumagai, Y.; Schmitz, D.A.; Singh, M.; Eiguren-Fernandez, A.; Froines, J.R. Redox activity of airborne particulate matter at different sites in the Los Angeles Basin. *Environ. Res.* **2005**, *99*, 40–47.
- [33] Gate, L.; Paul, J.; Nguyen Ba, G.; Tew, K.D.; Tapiero, H. Oxidative stress induced in pathologies: The role of antioxidants. *Biomed. Pharmacother.* **1999**, *53*, 169–180.
- [34] Thorn, K.A.; Arterburn, J.B.; Mikita, M.A. N-15 and C-13 NMR investigation of hydroxylamine -derivatized humic substances. *Environ. Sci. Technol.* **1992**, *26*, 107–116.
- [35] Ratasuk, N.; Nanny, M.A. Characterization and quantification of reversible redox sites in humic substances. *Environ. Sci. Technol.* **2007**, *41*, 7844–7850.
- [36] Klaus, V.; Hartmann, T.; Gambini, J.; Graf, P.; Stahl, W.; Hartwig, A.; Klotz, L.O. 1,4-Naphthoquinones as inducers of oxidative damage and stress signaling in HaCaT human keratinocytes. *Arch. Biochem. Biophys.* **2010**, *496*(2), 93–100.
- [37] Hou, T.; Zhang, X.; Xu, J.; Jian, C.; Huang, Z.; Ye, T.; Hu, K.; Zheng, M.; Gao, F.; Wang, X.; Cheng, H. Synergistic triggering of superoxide flashes by mitochondrial Ca²⁺ uniport and basal reactive oxygen species elevation. *J. Biol. Chem.* **2013**, *288*, 4602–4612.
- [38] Hansson, M.J.; Mansson, R.; Morota, S.; Uchino, H.; Kallur, T.; Sumi, T.; Ishii, N.; Shimazu, M.; Keep, M.F.; Jegorov, A.; Elmer, E. Calcium-induced generation of reactive oxygen species in brain mitochondria is mediated by permeability transition. *Free Rad. Biol. Med.* **2008**, *45*, 284–294.
- [39] Scarlett, J.L.; Sheard, P.W.; Hughes, G.; Ledgerwood, E.C.; Ku, H.H.; Murphy, M.P. Changes in mitochondrial membrane potential during staurosporine-induced apoptosis in Jurkat cells. *FEBS Lett.* **2000**, *475*, 267–272.
- [40] Hu, W. Anticancer therapy targeting the apoptotic pathway. *Lancet Oncol.* **2003**, *4*, 721–729.
- [41] Ly, J.D.; Grubb, D.R.; Lawen, A. The mitochondrial membrane potential (deltapsi(m)) in apoptosis; an update. *Apoptosis* **2003**, *8*, 115–128.
- [42] Schafer, Z.T.; Brugge, J.S. IL-6 involvement in epithelial cancers. *J. Clin. Invest.* **2007**, *117*, 3660–3663.
- [43] Fox, S.E.; Lu, W.; Maheshwari, A.; Christensen, R.D.; Calhoun, D.A. The effects and comparative differences of neutrophil specific chemokines on neutrophil chemotaxis of the neonate. *Cytokine* **2005**, *29*, 135–140.
- [44] Gangur, V.; Birmingham, N.P.; Thanavorakul, S. Chemokines in health and disease. *Vet. Immunol. Immunopathol.* **2002**, *86*, 127–136.
- [45] Iwanari, M.; Nakajima, M.; Kizu, R.; Hayakawa, K.; Yokoi, T. Induction of CYP1A1, CYP1A2, and CYP1B1 mRNAs by nitropolycyclic aromatic hydrocarbons in various human tissue-derived cells: Chemical-, cytochrome P450 isoform-, and cell-specific differences. *Arch. Toxicol.* **2002**, *76*, 287–298.
- [46] Choi, A.M.; Alam, J. Heme oxygenase-1: Function, regulation, and implication of a novel stress-inducible protein in oxidant-induced lung injury. *Am. J. Respir. Cell Mol. Biol.* **1996**, *15*, 9–19.
- [47] Ferrandiz, M.L.; Devesa, I. Inducers of heme oxygenase-1. *Curr. Pharm. Des.* **2008**, *14*, 473–486.